Node.js Introduction (Detailed Guide for PDF)

What is Node.js?

Node.js is an open-source, server-side JavaScript runtime environment that allows you to run JavaScript outside the browser.

History of Node.js

- Created by Ryan Dahl in 2009
- Built on Chrome's V8 engine
- Used for backend development

Why Use Node.js?

- Fast execution
- Non-blocking I/O
- Scalable applications
- Uses JavaScript on both frontend and backend

Features of Node.js

- Event-driven architecture
- Asynchronous programming
- Single-threaded
- High performance

Installing Node.js

Download from official website and install.

First Program

```
console.log("Hello Node.js");
```

Running Program

```
node filename.js
```

Modules

```
const fs = require('fs');
```

File System Module

```
const fs = require('fs');

fs.writeFileSync('file.txt', 'Hello');
```

HTTP Module

```
const http = require('http');

http.createServer((req, res) => {
  res.write('Hello World');
  res.end();
}).listen(3000);
```

NPM (Node Package Manager)

```
npm init
npm install express
```

Express.js (Framework)

```
const express = require('express');
const app = express();

app.get('/', (req, res) => {
  res.send('Hello Express');
});
```

```
app.listen(3000);
```

Event Loop

Handles asynchronous operations efficiently.

Asynchronous Programming

Callback

Promise

Async/Await

```
async function test() {
  return "Hello";
}
```

Example Project

```
const http = require('http');

http.createServer((req, res) => {
  res.write('Welcome to Node.js');
  res.end();
}).listen(3000);
```

Advantages of Node.js

- Fast and scalable
- Uses JavaScript everywhere
- Large ecosystem (npm)

Disadvantages of Node.js

- Not suitable for CPU-heavy tasks
- Callback complexity

Best Practices

- Use async/await
- Handle errors properly
- Keep code modular

Conclusion

Node.js is a powerful runtime for building fast and scalable server-side applications.

PDF Writing Tips

- Use headings
- Add examples
- Keep formatting clean